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PROJECT ON

Retail Banking Transaction Analysis

About me

Team Leader : Niharika Pentyala [B.Tech]

- Background - B.Tech Graduate
- Why you want to learn Data Science -

For me, Data Science is about understanding data and finding meaningful insights from it. I want to build strong skills in SQL, data analysis, and business analytics so that I can use data to solve real-world business problems and support better decision-making.

- Any work experience -

I am a fresher with no prior work experience. I am gaining practical experience through projects and training in Data Science and analytics.

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Project Overview

- Retail Banking Transaction Analysis is a MySQL Business Analytics project.
- The bank serves customers through different banking products such as accounts, cards, and loans.
- The bank operates through multiple branches and handles customer transactions and loan payments.
- MySQL is used to build the database and analyze the banking data.
- The analysis focuses on:
 - Customer profile & segmentation
 - Account usage & branch activity
 - Transaction patterns
 - Loan performance & repayment behaviour
 - Card usage & product engagement

Key Insight

Data analysis helps the Retail Banking team understand customer behaviour, banking activity, transaction patterns, loan performance, and product usage to support banking-related decisions.

Business Scenario

The bank wants to use data to understand its customers and banking activities and support better banking - related decisions.

Business Questions

- Which customer segments show different characteristics and behaviour?
- How does account activity vary across account types and branches?
- What transaction types and channels are most commonly used?
- Which loan types and branches show different repayment performance?
- How are customers using cards and other banking products?

Business Goal

Use MySQL-based analysis to identify customer, account, transaction, loan, and product-usage patterns and communicate meaningful business insights.

Business Objectives

1 Customer Analysis

Understand customer profiles and identify differences between customer segments.

2 Account & Branch Analysis

Understand account usage and compare activity across account types and branches.

3 Transaction Analysis

Identify transaction patterns and understand how money moves through the banking system.

4 Loan Performance

Evaluate loan performance and customer repayment behaviour.

5 Card & Product Engagement

Understand card usage and how customers use multiple banking products.

Dataset & Database Overview

Database Overview

The project contains 7 tables:

- **Customers** – customer details and segmentation
- **Accounts** – account information and balances
- **Branches** – branch details and locations
- **Loans** – loan details and loan status
- **Loan Payments** – loan repayment information
- **Cards** – card details and usage information
- **Transactions** – account transaction detail

Dataset Size

- **500 Customers**
- **700 Accounts**
- **40 Branches**
- **300 Loans**
- **2,000 Loan Payments**
- **600 Cards**
- **5,000 Transactions**

ER Diagram & Database Structure

Key Relationships

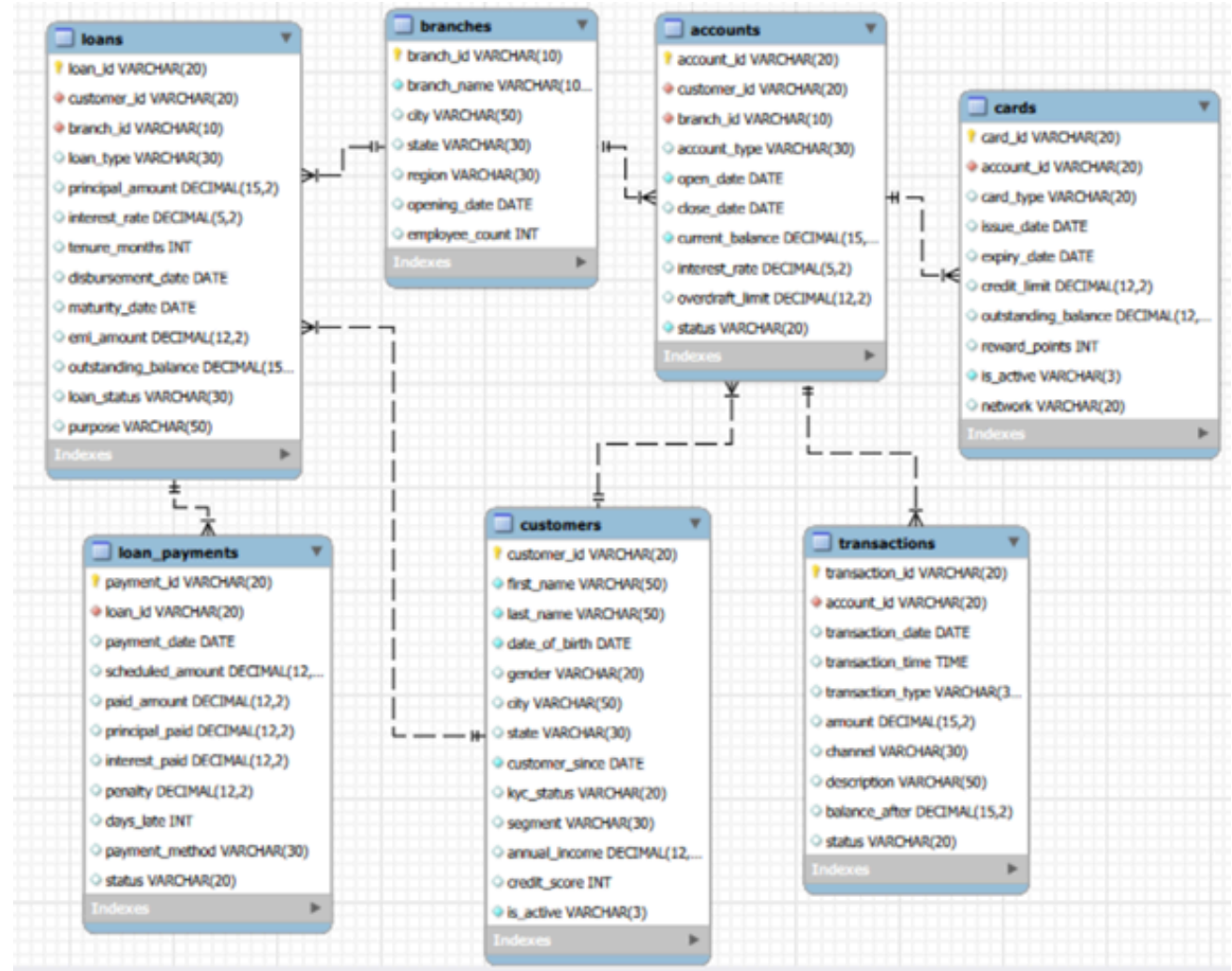
- Customers → Accounts
- Accounts → Transactions
- Accounts → Cards
- Customers → Loans → Loan Payments
- Branches → Accounts & Loans

Primary Keys

- customer_id
- account_id
- branch_id
- loan_id
- payment_id
- card_id
- transaction_id

Key Insight

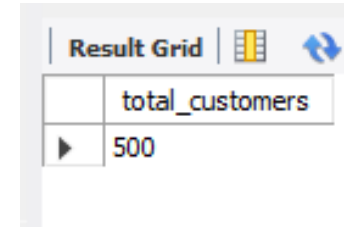
The database connects customers, accounts, transactions, cards, loans, loan payments, and branches, enabling analysis of overall banking activity.



Basic Analysis: Customers & Accounts

Q1. What is the total number of customers?

```
SELECT COUNT(*) AS total_customers  
FROM customers;
```

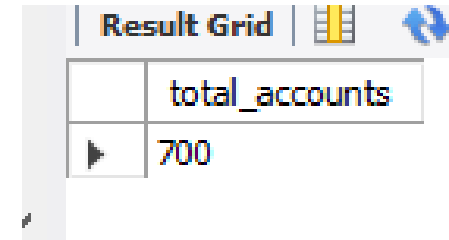


A screenshot of a SQL query result grid. The grid has two columns: 'total_customers' and a value '500'. The header row is labeled 'total_customers' and the data row shows the value '500'.

total_customers
500

Q2. What is the total number of accounts?

```
SELECT COUNT(*) AS total_accounts  
FROM accounts;
```

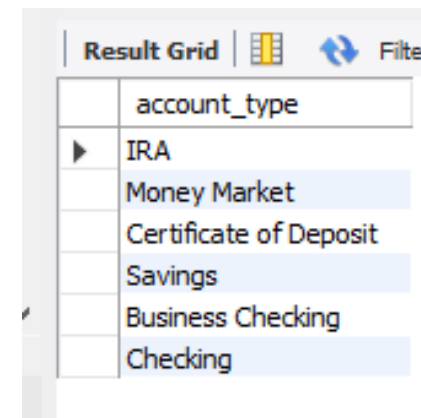


A screenshot of a SQL query result grid. The grid has two columns: 'total_accounts' and a value '700'. The header row is labeled 'total_accounts' and the data row shows the value '700'.

total_accounts
700

Q3. What are the different account types available?

```
SELECT DISTINCT account_type  
FROM accounts;
```



A screenshot of a SQL query result grid. The grid has two columns: 'account_type' and a list of account types. The header row is labeled 'account_type' and the data row shows the value 'IRA'. The data row is highlighted in blue.

account_type
IRA

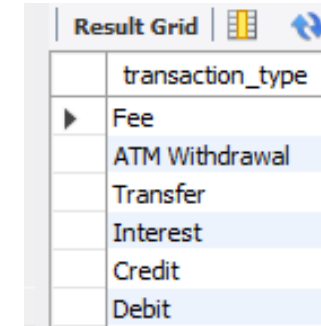
Key Insight

The bank has 500 customers and 700 accounts, indicating that some customers hold multiple accounts.

Basic Analysis: Transactions

Q. What are the different transaction types available?

```
SELECT DISTINCT transaction_type  
FROM transactions;
```

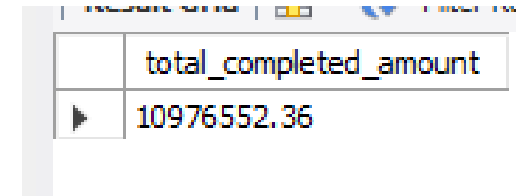


A screenshot of a database query result grid. The title bar says 'Result Grid'. The first row is the header 'transaction_type'. The subsequent rows are 'Fee', 'ATM Withdrawal', 'Transfer', 'Interest', 'Credit', and 'Debit', each with a small triangle icon to its left.

transaction_type
Fee
ATM Withdrawal
Transfer
Interest
Credit
Debit

Q. What is the total amount of completed transactions?

```
SELECT SUM(amount) AS total_completed_amount  
FROM transactions WHERE status = 'Completed';
```



A screenshot of a database query result grid. The title bar says 'Result Grid'. The first row is the header 'total_completed_amount'. The second row shows the value '10976552.36' with a small triangle icon to its left.

total_completed_amount
10976552.36

Key Insight

The bank handles different types of transactions, with a total completed transaction amount of approximately 10.98 million.

Basic Analysis: Loans Overview

Q. What is the total number of loans?

```
SELECT COUNT(*) AS total_loans  
FROM loans;
```

Result Grid	
	total_loans
▶	300

Q. What are the different loan types available?

```
SELECT DISTINCT loan_type  
FROM loans;
```

Result Grid	
	loan_type
▶	Business
	Credit Builder
	Personal
	Education
	Home
	Auto

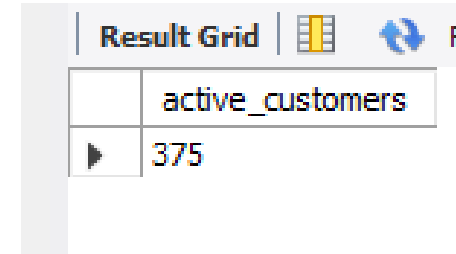
Key Insight

The bank has 300 loans across multiple loan types, showing a diverse loan portfolio.

Basic Analysis: Customers, Cards & Loans

Q. How many customers are currently active?

```
SELECT COUNT(*) AS active_customers  
FROM customers WHERE is_active = 'Yes';
```

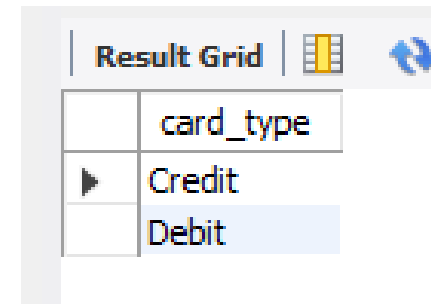


A screenshot of a database query result grid. The grid has a header row with the column name 'active_customers' and a data row with the value '375'. The interface includes a 'Result Grid' label, a grid icon, a refresh icon, and a 'Filter Row' label.

	active_customers
▶	375

Q. What are the different card types available?

```
SELECT DISTINCT card_type  
FROM cards;
```

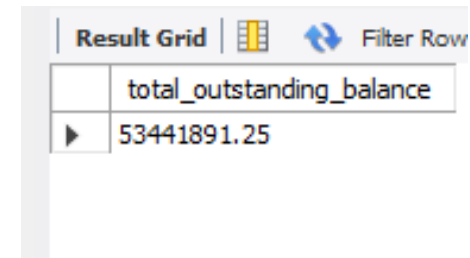


A screenshot of a database query result grid. The grid has a header row with the column name 'card_type' and a data row with the value 'Credit'. The interface includes a 'Result Grid' label, a grid icon, a refresh icon, and a 'Filter Row' label.

	card_type
▶	Credit
	Debit

Q. What is the total outstanding loan balance?

```
SELECT SUM(outstanding_balance) AS total_outstanding_balance  
FROM loans;
```



A screenshot of a database query result grid. The grid has a header row with the column name 'total_outstanding_balance' and a data row with the value '53441891.25'. The interface includes a 'Result Grid' label, a grid icon, a refresh icon, and a 'Filter Row' label.

	total_outstanding_balance
▶	53441891.25

Key Insight

The bank has 375 active customers, offers Credit and Debit cards, and has an outstanding loan balance of approximately 53.44 million.

Customer Profile & Segmentation Analysis

Key Findings

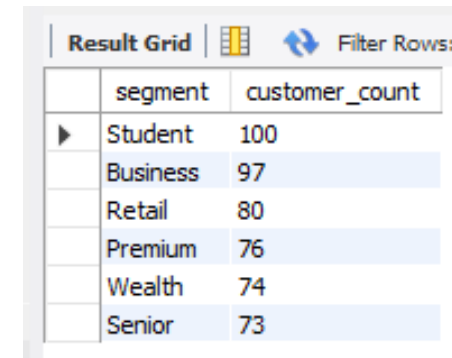
- Student segment has the highest number of customers (100).
- Business segment follows with 97 customers.
- Senior segment has the lowest number of customers (73).

Key Insight

- Customer distribution varies across segments.
- This helps the bank understand different customer groups and plan targeted banking strategies.

Q. How many customers are there in each customer segment?

```
SELECT segment,  
COUNT(*) AS customer_count  
FROM customers  
GROUP BY segment  
ORDER BY customer_count DESC;
```



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains the following data:

	segment	customer_count
▶	Student	100
	Business	97
	Retail	80
	Premium	76
	Wealth	74
	Senior	73

Customer Profile & Segmentation Analysis

Key Findings

- Premium customers have the highest average income (158,667.68).
- Retail customers have the highest average credit score (687.08).

Key Insight

- Customer segments differ in both size and financial characteristics.
- These differences can help the bank understand customer groups and plan targeted banking strategies.

Q. What is the average income and credit score across customer segments?

```
SELECT segment,  
ROUND(AVG(annual_income), 2) AS avg_income,  
ROUND(AVG(credit_score), 2) AS avg_credit_score  
FROM customers  
GROUP BY segment  
ORDER BY avg_income DESC;
```

Result Grid				Filter Rows:	
	segment	avg_income	avg_credit_score		
▶	Premium	158667.68	685.12		
	Student	158116.81	679.15		
	Senior	156745.53	684.32		
	Wealth	152389.12	679.89		
	Business	144267.52	683.95		
	Retail	140525.70	687.08		

Account Usage & Branch Activity Analysis

Key Findings

- Certificate of Deposit has the highest average balance (22,056.62).
- IRA has the second-highest average balance (21,279.47).
- Checking has the lowest average balance (3,963.92).

Key Insight

- Higher-balance account types hold more customer funds.
- This can help the bank understand account usage and improve product-focused strategies.

Q. Which account types have the highest average balance?

SELECT

account_type,

ROUND(AVG(current_balance), 2) AS avg_balance

FROM accounts

GROUP BY account_type

ORDER BY avg_balance DESC;

Result Grid			Filter Rows:
	account_type	avg_balance	
▶	Certificate of Deposit	22056.62	
	IRA	21279.47	
	Business Checking	17779.00	
	Money Market	15043.19	
	Savings	4924.70	
	Checking	3963.92	

Account Usage & Branch Activity Analysis

Key Findings

- 497 accounts are active, while 203 accounts are closed.

Key Insight

- Most accounts are currently active, indicating that the majority of accounts remain active.
- Account balance and status patterns can help the bank improve account management strategies.

Q. What is the number of active and closed accounts?

```
SELECT
    status,
    COUNT(*) AS account_count
FROM accounts
GROUP BY status
ORDER BY account_count DESC;
```

	Result Grid	Filter Row
	status	account_count
▶	Active	497
	Closed	203

Transaction Patterns Analysis

Key Findings

- Transfer has the highest transaction volume with 858 transactions.
- ATM Withdrawal follows with 857 transactions.
- Interest transactions recorded 855 transactions.
- Transfer has the highest total transaction amount at 5,933,692.01.
- Fee transactions have the lowest transaction volume with 784 transactions.

Key Insight

- Transfer and ATM Withdrawal are the most frequently occurring transaction types.
- The high transaction activity indicates strong usage of the bank's transaction services.

Q. Which transaction types have the highest transaction volume and total amount?

SELECT

transaction_type,

COUNT(*) AS transaction_count,

ROUND(SUM(amount), 2) AS total_amount

FROM transactions

GROUP BY transaction_type

ORDER BY transaction_count DESC;

Result Grid	Filter Rows:	Expc
transaction_type	transaction_count	total_amount
Transfer	858	5933692.01
ATM Withdrawal	857	5939130.04
Interest	855	6394799.91
Debit	828	5791548.90
Credit	818	6346746.51
Fee	784	5485289.92

Loan Performance & Repayment Analysis

Key Findings

- Education loans have the highest number of loans (64).
- Business loans have the highest total outstanding balance (10,056,247.55).
- Home loans have the second-highest total outstanding balance (8,364,122.07).
- Personal loans have the lowest total outstanding balance (7,676,485.94).

Key Insight

- Business loans represent the highest outstanding loan exposure.
- Education loans have the highest loan volume, while business loans have the highest outstanding balance.

Q. What is the loan count and total outstanding balance by loan type?

```
SELECT loan_type,  
       COUNT(*) AS loan_count,  
       ROUND(SUM(outstanding_balance), 2) AS  
total_outstanding  
FROM loans  
GROUP BY loan_type  
ORDER BY total_outstanding DESC;
```

Result Grid   Filter Rows:			
	loan_type	loan_count	total_outstanding
▶	Education	64	10937737.62
	Business	52	10056247.55
	Home	42	8364122.07
	Credit Builder	49	8300285.39
	Auto	48	8107012.68
	Personal	45	7676485.94

Card Usage & Product Engagement

Key Findings

- Debit cards are more common, with 358 cards.
- Credit cards account for 242 cards.
- The bank has 600 cards in total.
- Card usage can be further evaluated using card status, credit limits, outstanding balances, and reward points.

Key Insight

- Debit cards have higher representation than credit cards.
- Card-level analysis can help the bank understand product usage and customer engagement.

Q. What are the different card types and their total count?

```
SELECT  
    card_type,  
    COUNT(*) AS card_count  
FROM cards  
GROUP BY card_type  
ORDER BY card_count DESC;
```

	card_type	card_count
▶	Debit	358
	Credit	242

Overall Key Findings

Customers

- Student segment has the highest customer count.
- Premium customers have the highest average income.

Accounts

- Most accounts are active.
- Certificate of Deposit has the highest average balance.

Transactions

- Transfer has the highest transaction volume.
- Completed transactions total approximately 10.98 million.

Loans

- Education loans have the highest loan count.
- Business loans have the highest outstanding balance.

Cards

- Debit cards are more widely used than credit cards.
- Card analysis helps identify customer product preferences.

Business Recommendations

Business Recommendations

- Use customer segmentation to design targeted banking products and offers.
- Focus on high-balance account types to improve customer retention and product engagement.
- Monitor transaction patterns, especially high-volume transaction types such as transfers and ATM withdrawals.
- Closely monitor business loans because they have the highest outstanding loan exposure.
- Promote suitable card products based on customer needs and usage patterns.

Conclusion

- The project successfully analyzed customer, account, transaction, loan, and card data using MySQL.
- SQL queries helped identify important patterns and differences across banking activities.
- The analysis provided actionable insights into customer segments, account usage, transaction activity, loan exposure, and card usage.
- These insights can support better customer targeting, product management, and banking decisions.

THANK
YOU

